



ME 608: Thin Films And Devices

Course Code	ME 608
Course Name	Thin Films And Devices
Approval	8th Senate (Oct 2015)

Course Outline

Thin film science and technology have gone through a thorough development which results in numerous new devices (e.g, Light Emitting Diodes (LED), fuel cell and solar cell) and new materials with fundamentally new properties. Thin film research shares the knowledge from multi-disciplines (e.g., materials science, chemistry, solid state physics, mechanics and etc.) This undergraduate course is designed as an introductory class for those students who are interested in thin film fundamentals and processing for various industrial applications. A newly developed model called “The Art of Laying Apples” will be used throughout the course for explaining thin film concepts. Topics include, but are not limited to, fundamentals on crystal structures and defects in thin films, the basic nucleation and growth mechanisms of thin films (growth models, lattice matching epitaxy and domain matching epitaxy), thin film processing techniques (CVD, MOCVD, MBE, PLD, Laser-MBE, sputtering, and evaporation etc.), thin film growth instrumentation aspect (energy source, chamber configurations, vacuum systems and growth controllers), and several advanced topics related to electrical and optical devices. Lab or tour session(s) will be provided to promote teaching and learning. The following table provides a tentative guideline for course subjects